

Report: Hydraulic Press Timer Controls

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DIV:- B(B2)

Course :- Process Automation

Programming Language:- Ladder Diagram (LD) - Implemented in OpenPLC Editor

1. Introduction:-

A hydraulic press is a commonly used machine in industries such as sheet metal forming, automobile manufacturing, molding, and stamping. It operates using hydraulic cylinders to generate a compressive force. Traditionally, press operation required manual control by operators, which could lead to errors, fatigue, and safety hazards.

By automating the press cycle using PLC ladder logic, the machine operation becomes reliable, safe, and repeatable. In this project, we designed and simulated a Hydraulic Press Timer Control System using RSLogix/OpenPLC. The system ensures that the piston:

1. Waits briefly after the cycle begins.
2. Moves downward to press.
3. Holds the press for a set duration.
4. Retracts automatically to its original position.

This automation ensures both operator safety and production efficiency.

2. Objective:-

The objective of this project is to design a PLC ladder logic program to control a hydraulic press cycle using timers. The program must:

- Start and stop safely using push buttons.
- Control piston movement (forward, hold, retract) automatically.
- Include appropriate use of timers for sequential operations.
- Be validated through simulation in OpenPLC/RSLogix.

3. Input / Output List:-

The following table shows all system variables, their type, and their function:

Variable Name	Type	Description
Start	BOOL (Input)	Push button to initiate the press cycle.
Stop	BOOL (Input)	Normally Closed (NC) button to stop/reset the cycle.
M_0	BOOL (Internal)	Memory bit for latch (cycle holding).
TON0	TON (Timer)	5-second delay before piston forward motion.
TON1	TON (Timer)	7-second delay for holding the press.
TON2	TON (Timer)	10-second delay for piston retraction.
Press_A	BOOL (Output)	Extends the press piston (forward stroke).
Press_B	BOOL (Output)	Holds the piston in the pressing position.
Press_C	BOOL (Output)	Retracts the piston to the home position.

4. PLC Ladder Diagram and Program Explanation:-

The PLC ladder diagram is structured into four main rungs:

Rung 1: System Latching (Start/Stop):-

- The operator presses Start, which energizes the internal memory coil M_0.
- Stop button immediately resets the cycle.
- This latch ensures that the system runs continuously without holding the Start button.

Rung 2: Forward Motion Control:-

- When M_0 is active, the first timer TON0 (5s) is triggered.
- After 5 seconds, Press_A output energizes → piston moves forward.
- This controlled delay ensures the press does not start abruptly.

Rung 3: Holding the Press:-

- Once forward motion is complete, TON1 (7s) starts.
- Press_B energizes, keeping the piston in a downward position.
- The press remains in contact with the workpiece for exactly 7 seconds.

Rung 4: Retraction Control:-

- After the hold duration, TON2 (10s) activates.
- Press_C energizes, retracting the piston to its original position.
- At the end of the cycle, the press is ready for the next operation.

program0

Description:

Class Filter: All

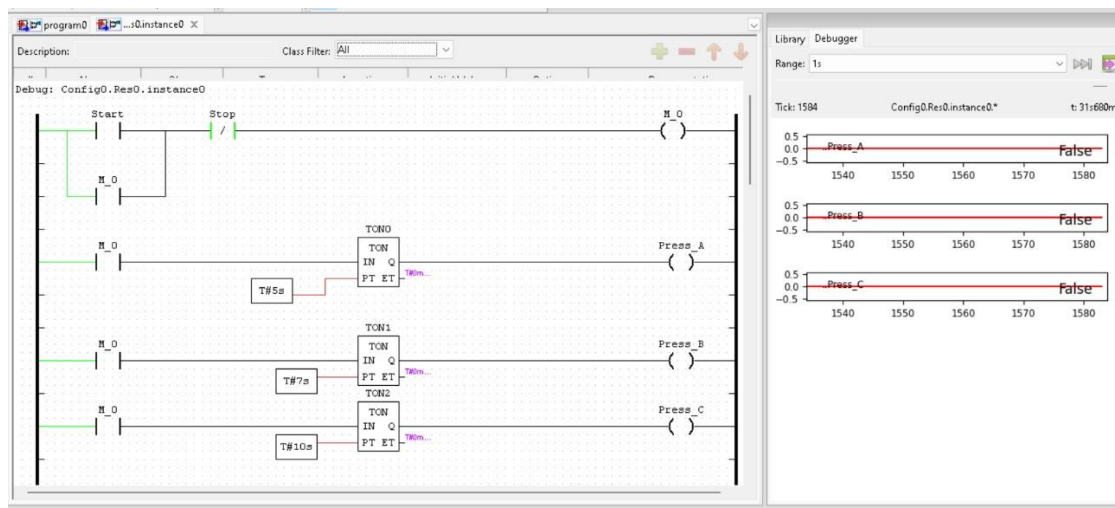
#	Name	Class	Type	Location	Initial Value	Option	Documentation
1	Start	Local	BOOL				
2	Stop	Local	BOOL				

The diagram shows a control system with three parallel branches. The first branch contains a normally open contact labeled 'Start' and a normally closed contact labeled 'M_0' in parallel, leading to a coil labeled 'M_0'. The second branch contains a normally open contact labeled 'M_0' leading to a TON timer block (TON0) with a preset time of T#5s, which is connected to a coil labeled 'Press_A'. The third branch contains a normally open contact labeled 'M_0' leading to a TON timer block (TON1) with a preset time of T#7s, which is connected to a coil labeled 'Press_B'. A fourth branch contains a normally open contact labeled 'M_0' leading to a TON timer block (TON2) with a preset time of T#10s, which is connected to a coil labeled 'Press_C'.

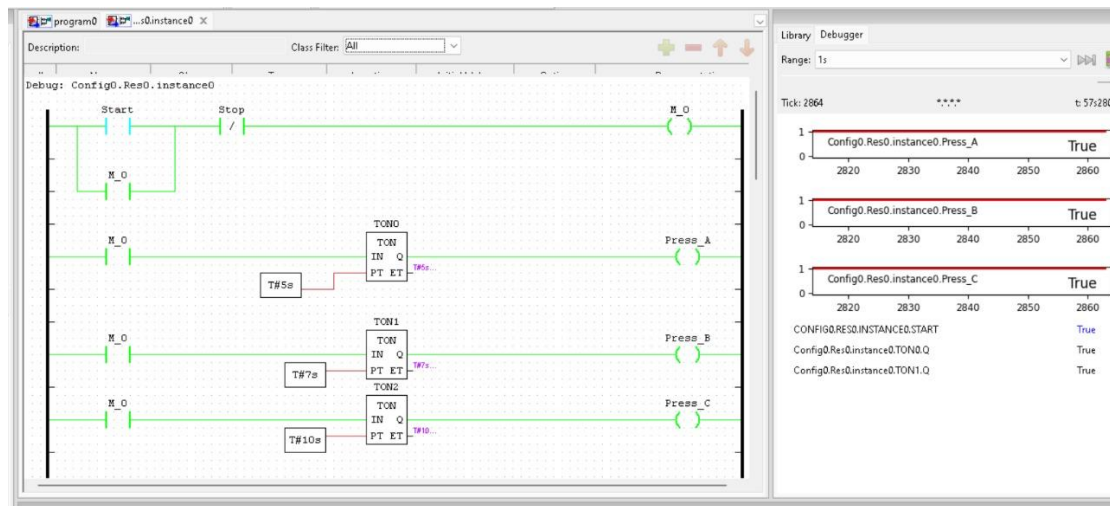
5. Simulation and Results:-

The program was implemented and tested in OpenPLC/RSLogix. The following sequence was observed:

- 1) Operator presses Start → Cycle begins (M_0 set).
- 2) After 5s, piston moves forward (Press_A = ON).
- 3) After 7s, piston holds position (Press_B = ON).
- 4) After 10s, piston retracts (Press_C = ON).
- 5) If Stop is pressed anytime, cycle resets and all outputs turn OFF.



program0							
Description:		Class Filter: All					
#	Name	Class	Type	Location	Initial Value	Option	Documentation
1	Start	Local	BOOL				
2	Stop	Local	BOOL				
3	M_0	Local	BOOL				
4	Press_A	Local	BOOL				
5	Press_B	Local	BOOL				
6	Press_C	Local	BOOL				
7	TON0	Local	TON				
8	TON1	Local	TON				
9	TON2	Local	TON				



6. Observations

- The use of timers ensures accurate cycle timing without operator intervention.
- The latch circuit provides a safe and reliable control method.
- The Stop button ensures emergency safety.
- Outputs (Press_A, Press_B, Press_C) are activated sequentially with no conflict.
- The system can be expanded with additional interlocks like limit switches and pressure sensors for real-world applications.

7. Conclusion:-

The Hydraulic Press Timer Control System was successfully designed and simulated using PLC ladder logic in RSLogix/OpenPLC. The project demonstrates:

- Correct use of timers to control sequential machine operations.
- Safe start/stop operation using a latch circuit.
- Smooth cycle transitions (forward → hold → retract).

- Improved safety, repeatability, and efficiency compared to manual operation.

This project can be extended to real-world hydraulic presses used in industries, ensuring controlled pressing operations in metal forming, molding, and stamping processes.